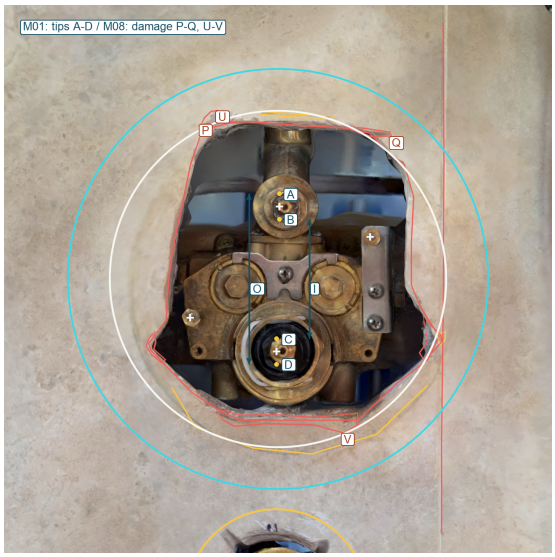
01 / Check the wall scale

Measurement worksheet | R004 | First checks

Calibrates the wall and valve scan; no CAD grid can establish physical scale.



With the trim removed only if safely accessible, touch the two visible brass stem tips lightly. Measure along the line joining the tips: outside span O, then inside gap I. The center spacing is (O + I) / 2, even if tip diameters differ. Repeat both readings. If access risks turning hardware, have the installer take the readings. A straight-on photograph with a ruler beside both tips is the fallback, with lower accuracy.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M01-O O: farthest tip edges A upper outside tangent to D lower outside tangent	unknown Useful +/-0.25	_____	_____	_____
M01-I I: gap between nearest tip edges B upper inside tangent to C lower inside tangent	unknown Useful +/-0.25	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

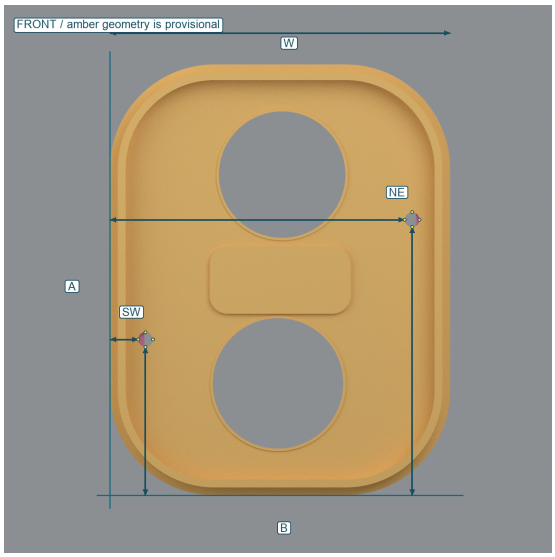
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

02 / Identify the inner plate datums

Measurement worksheet | R004 | First checks

Replaces the assumed outline and prevents misleading hole coordinates.



Place the loose factory inner plate on a padded surface. FRONT faces the handles (confirmed in the supplied photo); BACK faces the wall. Do not remove or compress the gasket. Use the leftmost tangent as datum A and bottommost tangent as datum B. Record the overall width and height; photograph front and back with those labels. The actual front photo shows a rounded rectangular outline; corner shape remains unmeasured. For a span beyond the caliper range, use two square blocks and a steel ruler.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M02-W Maximum metal outline width A left tangent to opposite right tangent	Recorded 130.35 Useful +/-0.5	_____	_____	_____
M02-H Maximum metal outline height B bottom tangent to top tangent	Recorded 165.1 Useful +/-0.5	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

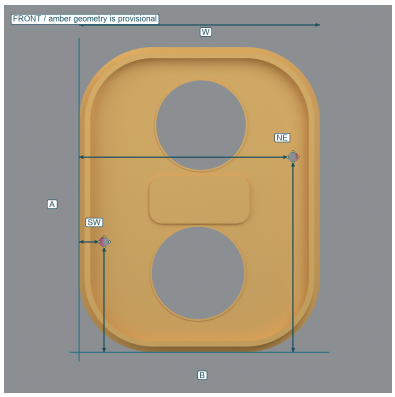
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

03 / Locate the two mounting openings

Measurement worksheet | R004 | First checks

Resolves hole/slot centers, size, and alignment with the actual diagonal studs.



Use the same A and B tangent datums as Check 02. Measure to both the near and far edges of each opening, not an eyeballed center. X center from A = (XN + XF)/2; Y center from B = (YN + YF)/2. Opening width = XF - XN; height = YF - YN. Do not use a curved edge as a moving datum: square blocks against the outline can transfer A/B to a ruler. If these are round holes or different slots, record that and photograph them. Take the plate off only using the applicable factory procedure.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M03-SW-XN SW opening: near X edge datum A to left edge of this opening	10.8 Useful +/-0.2	_____	_____	_____
M03-SW-XF SW opening: far X edge datum A to right edge of this opening	16.4 Useful +/-0.2	_____	_____	_____
M03-SW-YN SW opening: near Y edge datum B to bottom edge of this opening	56.9 Useful +/-0.2	_____	_____	_____
M03-SW-YF SW opening: far Y edge datum B to top edge of this opening	62.5 Useful +/-0.2	_____	_____	_____
M03-NE-XN NE opening: near X edge datum A to left edge of this opening	113.0 Useful +/-0.2	_____	_____	_____
M03-NE-XF NE opening: far X edge datum A to right edge of this opening	118.6 Useful +/-0.2	_____	_____	_____
M03-NE-YN NE opening: near Y edge datum B to bottom edge of this opening	102.8 Useful +/-0.2	_____	_____	_____
M03-NE-YF NE opening: far Y edge datum B to top edge of this opening	108.4 Useful +/-0.2	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

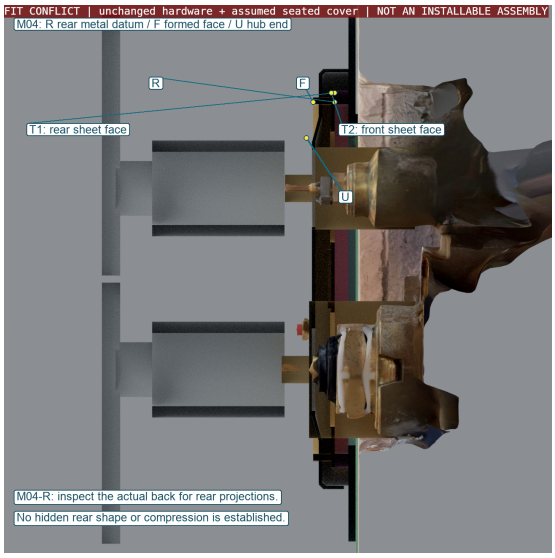
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

04 / Separate sheet thickness from formed depth

Measurement worksheet | R004 | First checks

Controls the stack and rear interference; thin sheet can have a much larger formed envelope.



Support the metal rear contact lands on equal blocks without loading the foam. Measure sheet thickness at an accessible unbent edge. Bridge the rear metal datum with a straightedge and measure to the frontmost formed metal face. Separately measure anything projecting toward the wall beyond that metal datum, and the forward hub projection. Mark the exact contact lands in the back photo. Do not flatten or bend the plate to measure it.

Already recorded: Overall plate and free foam depth: 15.44 mm. Free state; installed compression remains unknown.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M04-T Metal sheet thickness opposite metal faces at same accessible edge	1.5 Useful +/-0.1	_____	_____	_____
M04-D Rear metal datum to front formed face rear metal lands to forward formed face	9.4 Useful +/-0.2	_____	_____	_____
M04-R Largest rear metal projection rear land plane toward wall	unknown Useful +/-0.2	_____	_____	_____
M04-H Forward aperture rim / hub projection from rear datum rear metal datum to frontmost aperture rim or attached hub end	unknown Useful +/-0.2	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

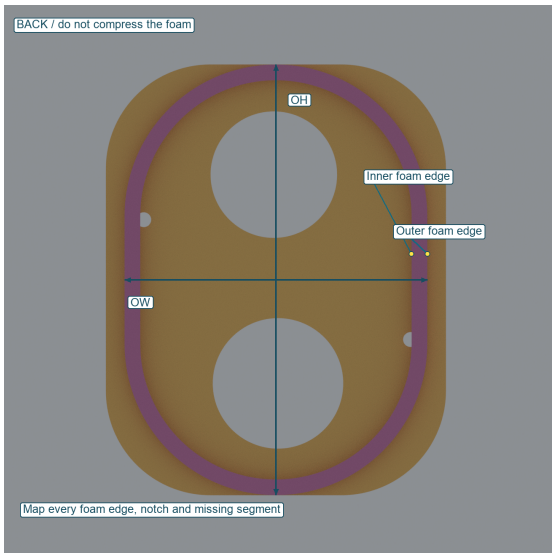
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

05 / Map the actual foam contact path

Measurement worksheet | R004 | First checks

Determines whether a plate aperture would remove gasket support.



Photograph the BACK face straight on, with A/B labels and a ruler alongside it. Mark the inner and outer foam edges and every notch, joint, tear or missing segment. Measure overall contact-path width/height, band width at left/right/top/bottom, and free thickness without squeezing the foam. Record the smallest band width below and the other readings in notes. A single width does not confirm a continuous seal; the photo/trace remains necessary.

Already recorded: Uncompressed foam thickness: approximately 6 mm; Representative foam band width: approximately 6 mm. Free state; installed compression remains unknown.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M05-OW Outer foam footprint width left to right outer foam edges	116.0 Useful +/-0.5	_____	_____	_____
M05-OH Outer foam footprint height bottom to top outer foam edges	165.0 Useful +/-0.5	_____	_____	_____
M05-W Smallest observed contact band width inner to outer foam edge at named location	6.0 Useful +/-0.5	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

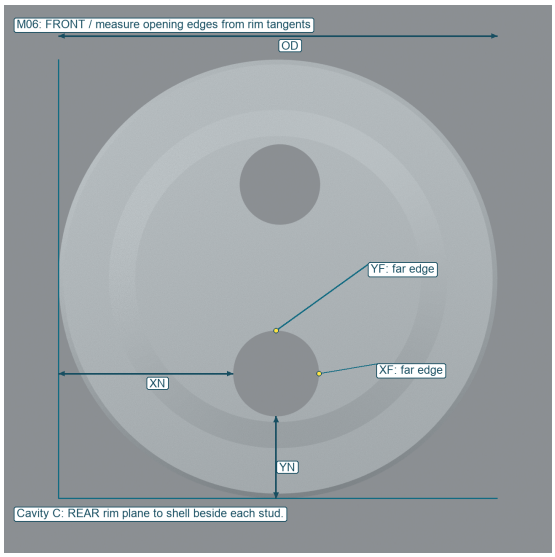
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

06 / Place the real outer trim

Measurement worksheet | R004 | First checks

Separates trim center from valve axes and mounting midpoint; replaces nominal diameter.



Protect the brushed finish. Measure the circular outer rim diameter in two directions and report any difference. Using left and bottom rim tangent blocks as datums, measure near/far edges of the LOWER control opening. Its center follows from the paired edges, so no center mark is needed. Measure the rear rim-contact plane to the inner cavity surface beside EACH stud location; report the smaller clearance and both locations in notes. Keep the drain notch at 6 oclock. Do not use the decorative front face as the rear contact datum.

Already reported: Outer rim diameter: 184.15 mm; Bottom rim tangent to lower opening near edge: reported 35.23 mm; contact clarification pending.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M06-XN Left rim tangent to lower opening near edge left tangent to left edge of lower bore	73.3 Useful +/-0.3	_____	_____	_____
M06-XF Left rim tangent to lower opening far edge left tangent to right edge of lower bore	109.3 Useful +/-0.3	_____	_____	_____
M06-YF Bottom rim tangent to lower opening far edge bottom tangent to top edge of lower bore	70.4 Useful +/-0.3	_____	_____	_____
M06-C Minimum rear cavity depth at stud locations rear rim plane to inside shell next to each stud	unknown Useful +/-0.25	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

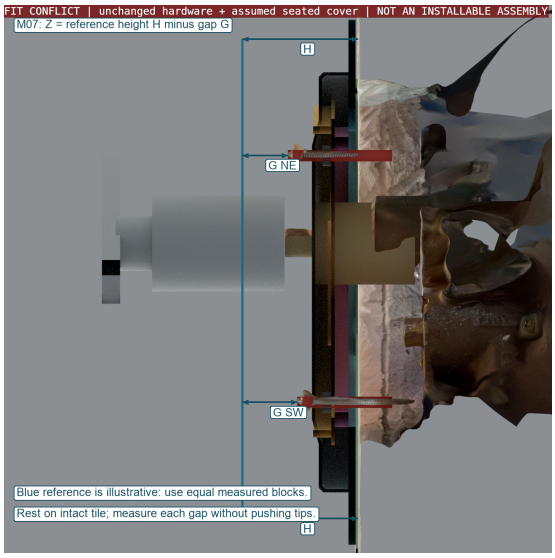
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

07 / Measure both stud projections

Measurement worksheet | R004 | First checks

Tests apparent seating interference without shortening studs or moving the valve.



Have the installer establish a straightedge plane on two equal-height blocks resting on intact tile. Choose blocks taller than both studs. Measure block height H, then gap G from the straightedge underside to each stud tip. Tip projection is H - G. Record any local tile lippage and do not rest the reference on a cut or grout recess. The installer also records adjustment range, engaged threads, nuts and adapters; visible tips cannot establish these. Do not cut studs or force the outer trim.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M07-H Reference-block height H tile-contact face to straightedge-contact face	unknown Useful +/-0.25	_____	_____	_____
M07-GSW Gap G to lower-left stud tip straightedge underside to SW tip	unknown Useful +/-0.25	_____	_____	_____
M07-GNE Gap G to upper-right stud tip straightedge underside to NE tip	unknown Useful +/-0.25	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

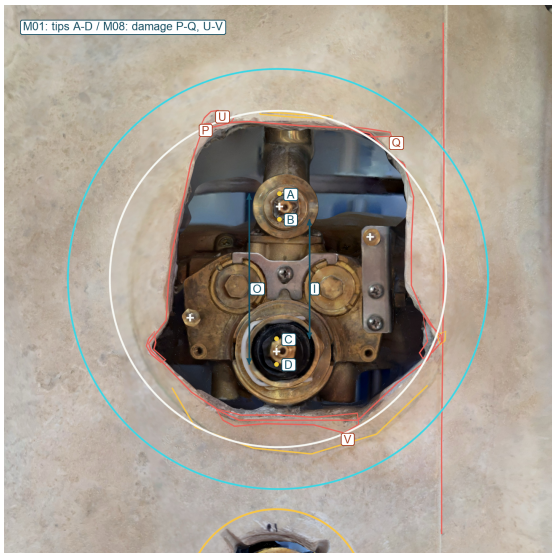
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

08 / Check the worst cuts and current state

Measurement worksheet | R004 | First checks

Independently checks scale/coverage and identifies changes since capture.



Measure the horizontal span P-Q from the leftmost upper cut end to the rightmost upper cut end, and the vertical span U-V from the highest upper damage to the lowest bottom chipped edge, using the labeled scan view. Do not add these as calibration inputs until their independent-check role is reviewed. Identify whether CUT-02/CUT-07 are real cuts or texture/shadow, and whether MARK-01 is only residue. At GROUT-01 record recess width/depth. No changes since capture were confirmed on 2026-10-14. Report only subsequent changes to piping, valve pose, studs or wall, with photographs of altered areas.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M08-X P-Q upper damage span leftmost CUT-01 end to farthest CUT-03 end	104.0 Useful +/-0.75	_____	_____	_____
M08-Y U-V damage height highest CUT-01 to lowest CUT-06	176.7 Useful +/-1	_____	_____	_____
M08-GW Adjacent grout recess width stone edge to stone edge at marked seam	unknown Useful +/-0.5	_____	_____	_____
M08-GD Adjacent grout recess depth intact stone plane to grout surface	unknown Useful +/-0.25	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

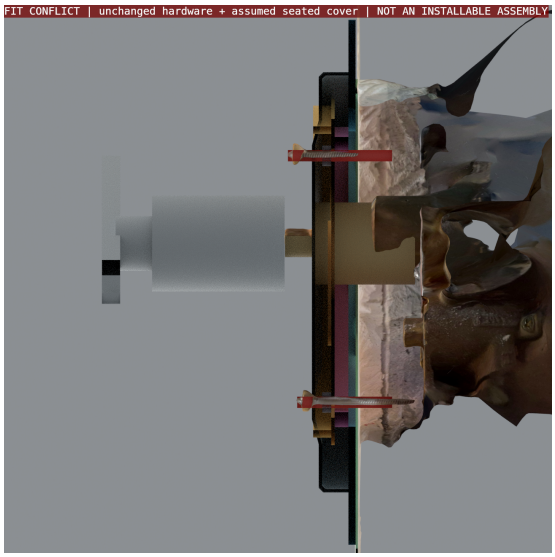
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

09 / Installer checks before selecting a repair

Measurement worksheet | R004 | Installer checks after first readings

Resolves inaccessible and hidden conditions without asking the homeowner to probe plumbing.



The user reports rear bolt contact with wood. Trace those fasteners to confirm whether they are the diagonal trim studs. Record whether the wood is structural framing or added blocking, rear clearance, usable threads, nut bearing seats, washers and stops. Trim studs thread into the valve; wood is not their required anchor. Separately identify the plaster guard, installed adapters and available alternatives; verify valve depth from the documented datum. Inspect foam compression, support and visible waterproofing through existing access. Keep cartridge/tool access and the lower diverter clear. The 7/16-inch limit applies to trim studs, not control stems. No stud cutting, wood drilling, valve move or unseen waterproofing detail is selected.

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

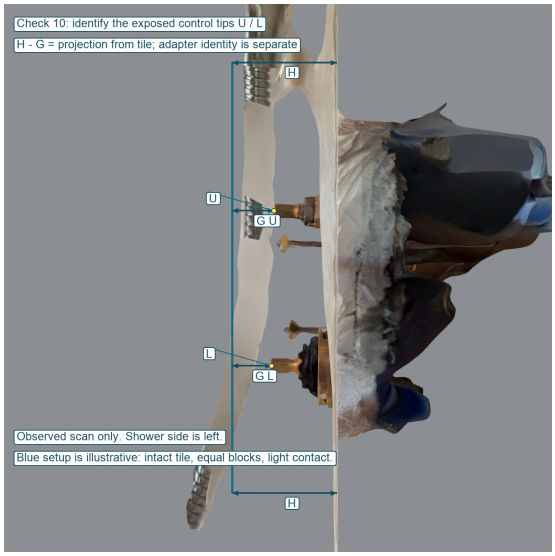
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

10 / Identify and measure the control tips

Measurement worksheet | R004 | Control-depth follow-up

Separates bare-stem projection, adapter length and valve rough-in depth; the upper scan tip currently triggers a conditional depth flag.



Identify the actual exposed upper tip U and lower tip L in a photo; record any attached spline/stem adapter, number or unknown identity. Do not remove a pressure-containing part to identify it. Use equal measured blocks on intact tile and a straightedge above both tips, as in Check 07. Record block height H and the gap from the straightedge UNDERSIDE to each as-found tip. Projection = H - G; a negative result means recessed behind tile. Read H again for this setup. These readings remain diagnostic until tip identity is resolved. The installer checks the plaster guard or manufacturer rough-in datum separately. Do not apply the 7/16-inch stud limit to these tips.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M10-H Control setup block height H tile-contact face to straightedge underside	unknown Useful +/-0.25	_____	_____	_____
M10-GU Gap G to upper as-found tip U straightedge underside to identified upper tip U	unknown Useful +/-0.25	_____	_____	_____
M10-GL Gap G to lower as-found tip L straightedge underside to identified lower tip L	unknown Useful +/-0.25	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

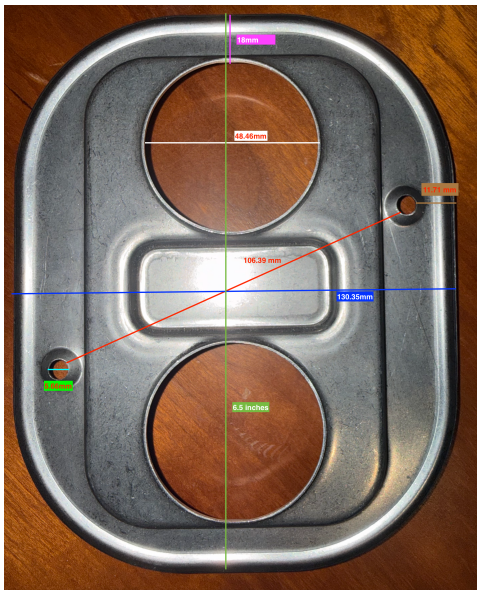
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

11 / Complete the photographed inner plate

Measurement worksheet | R004 | Actual plate follow-up

Uses the actual front face and separates edge gaps from center distances.



Front faces the handles. SW is the lower-left mounting hole; NE is the upper-right mounting hole. The supplied photo records width 130.35 mm, height 6.5 inches, upper opening 48.46 mm, top gap 18 mm, SW hole 5.66 mm, nearest-edge diagonal gap 106.39 mm and NE hole-to-right-rim gap 11.71 mm. Do not enter these again unless correcting them. Measure the NE hole and lower control opening independently. Diagonal center spacing requires BOTH mounting-hole diameters: gap + half of each diameter. Mounting-hole Y positions still need Check 03. The front photo does not establish the rear foam path or any Z depth.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M11-LD Lower control opening diameter left inside edge to right inside edge of lower large opening	unknown Useful +/-0.2	_____	_____	_____
M11-NED NE mounting-hole diameter opposite inside edges of upper-right round mounting hole, front view	unknown Useful +/-0.1	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

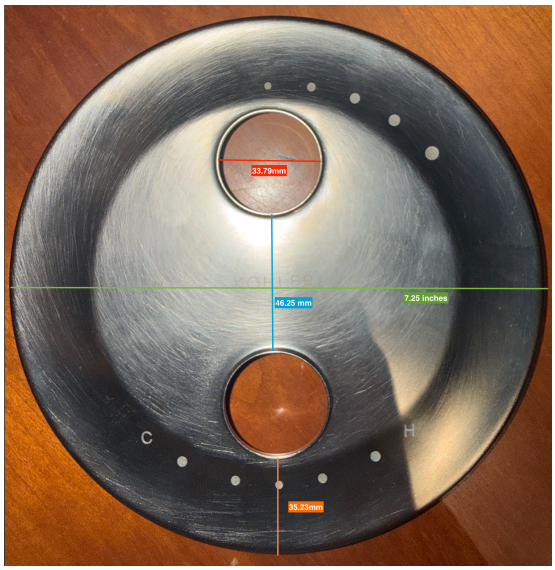
_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.

12 / Complete the photographed outer cover

Measurement worksheet | R004 | Actual plate follow-up

Separates measured opening diameters and edge gaps from cover placement and rear clearance.



The supplied photo shows the FRONT toward the handles. Outer diameter 7.25 inches (184.15 mm) was measured by tape; the upper opening is 33.79 mm by caliper. The blue annotation reports a 46.25 mm gap and the orange annotation 35.23 mm to the bottom rim; their exact inside-edge versus raised-lip contacts await clarification. Do not repeat these readings unless correcting them. Measure the LOWER opening inside diameter independently. Both diameters and a confirmed inside-edge gap are needed for vertical center spacing. The photo does not establish rear cavity depth: use Check 06-C beside both studs.

ID / contact	Estimate mm	Raw reading	Unit	Repeat readings
M12-LD Outer cover: lower opening inside diameter left inside edge to right inside edge of lower cover opening	unknown Useful +/-0.2	_____	_____	_____

Date: _____ Tool / method: _____

Photo / trace reference: _____

Notes, repeatability, condition and uncertainty: _____

_____

Amber model geometry is assumed. Do not use estimates as your readings. Return raw values and units; do not round to match the model.